

NarraLens

AI-Generated Data Report

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200 rows × 7 columns

Key Insights

↑ Total Records

200

Patient records analyzed across 7 clinical variables

■ Avg. Days Hospitalized

15.7 days

Mean inpatient stay duration (range: 1–29 days)

■ Anomalies Flagged

10 (5%)

Records flagged by IsolationForest across clinical variables

■ Missing Values

0

Zero empty cells across all 1,400 data points

★ Top Readmission Predictor

$r = 0.314$

NumMedications vs. Readmitted — strongest correlation in dataset

■ Avg. Medications per Patient

9.5 meds

Mean NumMedications (range: 0–19, CV = 60.51%)

Statistical Summary

Column	Mean	Median	Std Dev	Min	Max	Missing%
Age	52.848	53.6	21.232	18.4	89.06	0.0%
BloodPressure	120.525	125	35.16	60.61	178.86	0.0%

Column	Mean	Median	Std Dev	Min	Max	Missing%
NumMedications	9.495	10	5.745	0	19	0.0%
DaysHospitalized	15.715	16	8.309	1	29	0.0%
NumDiagnoses	5.055	5	2.541	1	9	0.0%
PrevAdmissions	1.925	2	1.414	0	4	0.0%
Readmitted	0.75	1	0.434	0	1	0.0%

Top Correlations

Column A	Column B	Correlation	Strength
NumMedications	Readmitted	0.3138	Weak
Age	Readmitted	0.1927	Weak
BloodPressure	PrevAdmissions	0.1390	Weak
DaysHospitalized	NumDiagnoses	-0.1354	Weak
NumDiagnoses	Readmitted	0.1128	Weak
BloodPressure	Readmitted	0.0869	Weak
PrevAdmissions	Readmitted	0.0839	Weak
NumMedications	NumDiagnoses	0.0732	Weak

AI-Generated Narrative Report

Executive Summary

This dataset captures 200 hospital patient records across seven clinical variables, revealing that the average patient is approximately 53 years old, spends nearly 16 days hospitalized, and carries close to 10 active medications. The most striking signal is that the number of medications prescribed is the strongest predictor of readmission ($r = 0.31$), while 10 anomalous records — likely representing medically complex or atypical cases — warrant individual clinical review.

Dataset Overview

The dataset comprises 200 rows and 7 numeric columns covering patient demographics (Age), clinical indicators (BloodPressure, NumDiagnoses), treatment metrics (NumMedications, DaysHospitalized), and hospitalization history (PrevAdmissions, Readmitted). There are no categorical or datetime columns, which limits segmentation analysis but simplifies numeric modeling. Data quality is excellent — zero missing values across all 1,400 cells — making this dataset immediately ready for predictive modeling without imputation. Ten anomalies were flagged using IsolationForest, representing 5% of the patient population, a meaningful but manageable rate for a clinical dataset.

Key Findings

- **Medication Load Is the Strongest Readmission Signal:** Patients prescribed more medications show the highest correlation with readmission ($r = 0.314$). With a mean of 9.5 medications per patient and a range spanning 0 to 19, polypharmacy appears to be a meaningful risk stratifier — patients near the upper end of this range likely represent the highest readmission risk cohort.
- **Age Skews Toward a Younger Distribution Despite High Mean:** Although the mean age is 52.8 years, 68.5% of patients fall above the mean, and the coefficient of variation is 40.18%, indicating meaningful spread across the full adult lifespan (18.4–89.1 years). This suggests the dataset captures a genuinely broad patient population rather than a specialized elderly cohort.
- **Hospitalization Duration Is Remarkably Long on Average:** The mean stay of 15.7 days (median: 16 days) with a range of 1–29 days indicates a predominantly inpatient, complex-care population. A coefficient of variation of 52.87% reflects substantial variability — some patients are discharged within days while others approach a full month of care.
- **Blood Pressure Spans a Dangerously Wide Range:** BloodPressure ranges from 60.61 to 178.86 mmHg, with a mean of 120.5 and a standard deviation of 35.2. Nearly half (48.4%) of patients fall below the mean, and the spread covers both hypotensive and severely hypertensive territory, suggesting this variable may identify distinct patient subgroups with different cardiovascular risk profiles.
- **Prior Admissions Are Uniformly Low but Clinically Relevant:** PrevAdmissions averages just 1.93 (median: 2.0, max: 4), meaning most patients have limited prior hospitalization history. However, the positive correlation with BloodPressure ($r = 0.139$) hints that patients with chronically elevated blood

pressure may be accumulating repeat admissions over time, a pattern worth monitoring prospectively.

Statistical Deep Dive

Age is broadly distributed (std = 21.2 years, CV = 40.18%), spanning nearly the entire adult lifespan. The close proximity of mean (52.8) and median (53.6) suggests a roughly symmetric, near-normal distribution without severe skew, though the 68.5% of patients above the mean warrants further histogram inspection for potential left-tail thinness in younger patients.

BloodPressure has a mean of 120.5 mmHg and a standard deviation of 35.2 (CV = 29.17%). The median of 125.0 sits slightly above the mean, indicating mild left skew — a cluster of hypotensive patients pulls the mean downward. The range of 60.61–178.86 is clinically extreme, suggesting the dataset may capture patients in acute distress at both ends.

NumMedications is the most variable clinical metric (CV = 60.51%), ranging from 0 to 19 with a mean of 9.5 and median of 10. The near-perfect alignment of mean and median confirms a symmetric distribution, but the high spread reflects genuinely divergent medication burdens across the patient population.

DaysHospitalized shows a high CV of 52.87%, with 84.5% of patients falling above the mean of 15.7 days — a statistical artifact indicating that a subset of very short-stay patients (near the 1-day minimum) pulls the mean below what most patients experience. The median of 16 days is likely the more representative central value.

NumDiagnoses is tightly distributed (mean = 5.06, std = 2.54, range = 1–9), reflecting a capped diagnostic classification structure. **PrevAdmissions** is similarly constrained (mean = 1.93, std = 1.41, max = 4), consistent with a dataset representing relatively early-stage chronic patients rather than frequent-flyer hospital users.

Correlations & Relationships

The correlation matrix reveals a dataset where relationships are real but modest — no pair of variables shares a strong ($r > 0.5$) relationship, indicating that readmission risk is multifactorial rather than driven by a single dominant variable.

NumMedications → **Readmitted** ($r = 0.314$, weak positive) is the headline relationship. While statistically weak, in a clinical context this is practically meaningful: higher medication counts reflect

greater medical complexity, more comorbidities, and greater discharge fragility. This variable should be a primary feature in any readmission prediction model.

Age → **Readmitted** ($r = 0.193$, negligible positive) confirms the intuitive expectation that older patients are marginally more likely to be readmitted, but the effect is smaller than expected, suggesting that age alone is an insufficient risk flag without supporting clinical context.

BloodPressure → **PrevAdmissions** ($r = 0.139$, negligible positive) is subtle but worth noting: patients with higher blood pressure tend to have slightly more prior admissions. This may reflect an uncontrolled hypertension pathway — patients who fail to manage blood pressure are recycled through the hospital system more frequently.

DaysHospitalized → **NumDiagnoses** ($r = -0.135$, negligible negative) is counterintuitive. One might expect more diagnoses to drive longer stays, but the slight negative correlation may reflect that highly diagnostic patients receive faster, protocol-driven care while less-diagnosed patients linger due to diagnostic uncertainty. This relationship merits deeper qualitative investigation.

Anomalies & Outliers

IsolationForest flagged **10 records** as anomalous, representing **5.0% of the dataset** — a clinically meaningful proportion. In a dataset with no missing values and well-bounded ranges, these anomalies likely represent multivariate outliers: patients who are unusual not because of a single extreme value, but because their combination of variables (e.g., very young age with very high medication counts, or very short hospitalization despite many diagnoses) defies the typical patient profile.

Key columns most susceptible to anomalous extremes include **BloodPressure** (range: 60.61–178.86, std = 35.2), **NumMedications** (range: 0–19, CV = 60.51%), and **DaysHospitalized** (range: 1–29, CV = 52.87%). These three columns exhibit the widest relative spread and are the most likely contributors to anomaly flags.

Clinically, these 10 records may represent: patients in acute crisis at admission, data entry errors in the source system, or genuinely rare clinical presentations. They should be reviewed by a clinician before being included in a predictive model, as they can disproportionately influence regression coefficients and classification boundaries.

Trends & Patterns

Without temporal or categorical columns, structural patterns emerge from distributional shapes and variable interactions.

Concentration around middle-age hospitalization: The age distribution (mean $\approx$ 53, spanning 18–89) combined with the long average stay (15.7 days) suggests the core patient population represents middle-aged adults with multi-system chronic disease — not primarily geriatric or pediatric populations.

High medication burden across the board: With 84.5% of hospitalization days above the mean and a mean of $\sim$ 10 medications per patient, this dataset captures a population that is medically complex at baseline. The near-zero minimum (0 medications) and high CV (60.51%) suggest a bimodal structure may exist — patients with minimal pharmaceutical management versus those in intensive polypharmacy regimens.

Blood pressure asymmetry: The median (125.0) exceeding the mean (120.5) by 4.5 points, combined with 48.4% of patients below the mean, indicates a concentration of readings in the normal-to-elevated range with a left tail of hypotensive outliers. This could reflect patients presenting in shock or post-surgical states.

NumDiagnoses as a natural ceiling variable: With a max of 9 and mean of 5.06, this variable appears structurally capped — likely reflecting a standardized coding schema (ICD categories) rather than a true clinical maximum. Its tight standard deviation (2.54) compared to other variables makes it the most stable predictor in the dataset.

Recommendations

- **Build a Readmission Risk Scoring Model Anchored on NumMedications:** Given that NumMedications holds the strongest correlation with readmission ($r = 0.314$), develop a logistic regression or gradient-boosted classifier using all available variables with NumMedications as a primary feature. Even modest predictive lift ($AUC > 0.65$) could enable targeted discharge planning for high-risk patients before they leave the hospital.
- **Investigate and Adjudicate the 10 Anomalous Records:** Commission a clinical data governance review of the 10 flagged anomalies. Determine whether they represent legitimate edge cases (include with flagging), data entry errors (correct or exclude), or rare clinical phenotypes (retain as a separate analytical stratum). Resolving these records will improve model reliability and statistical integrity.
- **Segment Patients by Medication Burden for Targeted Interventions:** Create three medication-load tiers — low (0–6), moderate (7–13), and high (14–19) — and analyze readmission rates, blood pressure control, and hospitalization length within each tier. Patients in the high-medication tier likely require enhanced post-discharge pharmacy coordination, home health

referrals, or medication reconciliation programs.

- **Enrich the Dataset with Categorical and Temporal Variables:** The absence of categorical columns (e.g., diagnosis category, discharge disposition, insurance type) and datetime columns (e.g., admission date, discharge date) severely limits segmentation and trend analysis. Prioritize a data pipeline enhancement to capture at minimum: primary diagnosis group, discharge destination, and admission timestamp. These additions would unlock time-series readmission rate tracking and demographic subgroup analysis.

Conclusion

This 200-patient clinical dataset is analytically clean and structurally sound, offering a reliable foundation for readmission risk modeling despite the absence of categorical and temporal enrichment. The convergence of high medication burden, extended hospitalization, and modest but consistent age-related risk signals points toward a medically complex patient population where targeted, data-driven discharge planning could meaningfully reduce preventable readmissions.